TWISTED CONJUGACY IN SPHERICAL ARTIN GROUPS AND THEIR CRYSTALLOGRAPHIC QUOTIENTS

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Abstract. Every irreducible Artin group of spherical type and rank at least two has property R_∞. We complete this classification by treating D_5, E_6, E_7, E_8, F_4, H_3 and H_4. We also classify the crystallographic quotients $A/[P,P]$, where P is the pure subgroup: they have R_∞ in every rank at least three, and in rank two, of type $I_2(m)$, exactly when $4 \mid m$ or when m is odd and $3 \mid m$. Thus this quotient construction does not preserve R_∞ in general. In type $I_2(5)$ we construct an automorphism of $A/[P,P]$ with Reidemeister number 18.

The proof uses the characteristic pure subgroup and its abelianization, viewed as the permutation lattice on the reflections. Rational, absolutely irreducible constituents of multiplicity one yield a fixed vector after composing with a suitable inner automorphism. The finite action that must be controlled is governed by automorphisms of Coxeter groups, so no description of $\mathrm{Aut}(A)$ is required. The constituents are explicit: eigenspaces of strongly regular graphs for E_n, frame modules for F_4 and H_3, and matrix coefficients in the icosian model for H_4.

The contrast between the positive quotients and the dihedral example is governed by the rational holonomy representation and the integral normaliser compatible with the crystallographic extension.

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1. Introduction

Let G be a group and $\varphi \in \mathrm{Aut}(G)$. Elements $x, y \in G$ are *φ-twisted conjugate* if $y = zx\varphi(z)^{-1}$ for some $z \in G$. The number $R(\varphi) \in \{1, 2, \ldots, \infty\}$ of classes is the *Reidemeister number*, and G has *property* R_∞ if $R(\varphi) = \infty$ for every $\varphi \in \mathrm{Aut}(G)$ [18, 16].

For an Artin group $A = A(\Gamma)$ with Coxeter group $W = W(\Gamma)$, the question "which Artin groups have R_∞?" is Problem 8.6 of the AIM problem list on Artin groups [1]. Known cases include:

- braid groups [15] and pure braid groups [9];
- the spherical types A_n, B_n, D_4, $I_2(m)$ [5];
- D_n for $n \geq 6$ [35, Thm. 1];
- many non-spherical types [27, 35, 34].

The case D_5 is left open (as a conjecture) in [35, §1]. Paris–Soroko computed $\mathrm{Aut}(A[D_5])$ [33, Thm. 1], from which R_∞ follows by a short argument (Remark 5.3); [33] does not state it. The list in [34, §1] includes D_5, attributing it to [35]. We include an independent proof for D_5 in the completion below. For spherical types the known proofs use an explicit description of $\mathrm{Aut}(A)$ or $\mathrm{Aut}(A/Z(A))$. For $I_2(m)$ they use instead the virtual freeness of $A/Z(A)$ [5].

Theorem A. *Every irreducible Artin group of spherical type and rank at least two has property R_∞.*

The cases proved here are $D_5, E_6, E_7, E_8, F_4, H_3, H_4$; see Section 4 and Proposition 5.2. Together with [5] (types $A_n, B_n, D_4, I_2(m)$) and [35, Thm. 1] (D_n, $n \geq 6$), they give Theorem A. Rank one

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must be excluded: $A(A_1) \cong \mathbb{Z}$, and inversion has Reidemeister number 2. The method also gives uniform proofs in types A_n, B_n, D_n (Section 5).

Let $P = \ker(A \to W)$. Marin [31, Thm. 2.1] proves that $A/[P,P]$ is crystallographic with holonomy $W/Z(W)$. Our second result concerns all automorphisms of this quotient, including those that need not lift to A.

Theorem B. *Let X be irreducible spherical of rank at least two, and let $\Gamma_X = A(X)/[P,P]$. If X has rank at least three, then Γ_X has property R_∞. In rank two, writing $X = I_2(m)$ with $m \geq 3$, we have*

$$\Gamma_X \text{ has } R_\infty \quad \Longleftrightarrow \quad 4 \mid m \quad \text{or} \quad (m \text{ odd and } 3 \mid m).$$

The proof is given in Section 6. For the exceptional types it uses the full holonomy-automorphism calculations in Appendix B; the classical and dihedral families have uniform proofs. Since P and hence $[P,P]$ are characteristic [7], the positive cases of Theorem B imply R_∞ for the corresponding Artin groups. Together with the known dihedral Artin-group result [5], this also proves Theorem A, without the Franzsen–Howlett automorphism decomposition. The next result makes the failure of preservation concrete.

Proposition C. *The crystallographic quotient $A(I_2(5))/[P,P]$ admits an automorphism Φ with $R(\Phi) = 18$. In particular, it does not have property R_∞.*

Thus $A(I_2(5))$ has R_∞, whereas its natural crystallographic quotient does not. Section 6.7 gives the affine automorphism, checks the precise translation lattice and extension, and counts twisted classes directly. The count takes account of torsion. The positive results use absolutely irreducible multiplicity-one constituents to restrict commuting integral maps to signs. In the dihedral example the nontrivial rational constituent instead has an infinite-order unit in its commutant; a separate extension calculation shows that this unit yields an admissible automorphism.

A close precedent is the 2024 work of Dekimpe–Gonçalves–Ocampo [10, Thms. 28, 31], which proves R_∞ for $VB_n/[VP_n, VP_n]$ for $n = 3, 4$. Those are virtual-braid quotients; the spherical Artin quotients in Theorem B have different holonomy modules and extensions. Recent work also studies torus bundles arising from Bieberbach subgroups of classical braid quotients [30] and crystallographic quotients of framed braid groups [29]. Ocampo points to Nielsen and Reidemeister numbers of fibre-preserving maps as further questions for the torus bundles [30, §1]. The smallest positive case A_2 is already implicit in earlier classifications: its quotient is space group IT 161 [23, §5], which is absent from the complete list of three-dimensional groups without R_∞ in [11, Table 2].

Significance and novelty. For the six exceptional types $\mathrm{Aut}(A)$ is not known (cf. [33, §1]). The general criteria based on hyperbolic actions of $\mathrm{Aut}(A)$, or on Aut-invariant quasimorphisms [19, 20, 36], also require control of $\mathrm{Aut}(A)$.

Theorem A would follow from Conjecture (NA) of [35, §6]: finitely generated, residually finite, non-amenable groups have R_∞. This appears as a theorem in [17], but its proof is reported to be flawed [35, §6]. Our results verify (NA), and Conjecture (R) [28, Problem 19.28], for these groups.

The broad strategy of obtaining an eigenvalue one via representation theory has precedents:

- Dekimpe–Gonçalves–Ocampo [9], for pure braid groups;
- Dekimpe–De Rock–Penninckx [8, Thm. 4.7], for flat manifolds. They require a $\mathbb{Q}$-irreducible $\mathbb{Z}$-subrepresentation η_0 of multiplicity one in the holonomy representation whose image is not $\mathbb{Q}$-conjugate to that of an inequivalent $\mathbb{Q}$-subrepresentation, and such that for every D_0 in the integral normaliser of $\eta_0(F)$ some $\eta_0(A)D_0$, $A \in F$, has eigenvalue one;
- the determinant criterion of Dekimpe–Penninckx for crystallographic groups [12, Cor. 3.10], as stated in [11, Thm. 3.3];
- the eigenvalue-one property of Hiss–Lutowski [25, 26] (announced in [24]): for every finite group G, every non-trivial irreducible $\mathbb{R}G$-module V of odd dimension and every $n \in N_{GL(V)}(\rho(G))$ of finite order, some $\rho(g)n$ has eigenvalue 1 [25, Def. 1.1.3, Thm. 1.1.5].

The proof uses the classification of finite simple groups. Section 6.4 verifies the hypotheses for five cases of Theorem B.

What is new here is:

- the application to Artin groups through the *characteristic reflection lattice* $H_1(P;\mathbb{Z}) \cong \mathbb{Z}^{\mathcal{T}}$;
- the control of the relevant finite symmetries by $\mathrm{Aut}(W)$ [22];
- the explicit sign and swap witnesses;
- the spherical Artin-group classification, the classification of their crystallographic quotients, and the explicit dihedral counterexample to preservation under $A \mapsto A/[P,P]$.

The explicit witness argument avoids the classification of finite simple groups. It still uses results on finite Coxeter groups and exact exhaustive computations; we do not claim an elementary proof of all its inputs.

Overview of the Artin-group argument. Let $P = \ker(A \to W)$, let $\mathcal{T}$ be the set of reflections, and let $\rho\colon W \to \mathrm{Sym}(\mathcal{T})$ be the conjugation action.

- P is characteristic, and $H_1(P;\mathbb{Z}) \cong \mathbb{Z}^{\mathcal{T}}$ is a permutation module (Section 2).
- An automorphism φ acts on $\mathbb{Z}^{\mathcal{T}}$ as πN, where π realises the induced automorphism of $\rho(W) \cong W/Z(W)$ and N commutes with $\rho(W)$. On a good constituent U, $N = \pm 1$.
- For a suitable inner automorphism ι_a, the composite $\varphi' = \iota_a \circ \varphi$ has a fixed vector. So $R(\varphi'|_P) = \infty$, hence $R(\varphi') = \infty$, and $R(\varphi) = R(\varphi')$ (Section 3).
- The possible π are controlled by $\mathrm{Aut}(W)$. Explicit constituents and witnesses finish the proof (Section 4).

Formal verification. A Lean 4/Mathlib formalisation accompanies this paper. It contains no `sorry`, no `axiom` and no `native_decide`. The Lean kernel checks:

- the Reidemeister lemmas and the criterion (both cases);
- the Artin group as a presented group, the map f, and the equivariant surjection $P^{ab} \to \mathbb{Z}^{\mathcal{T}}$;
- for each exceptional type and for D_5, finite certificates (commutant-scalar constituents and ± 1 witnesses);
- the finite part of Lemma 4.1: by a search-tree (orbit–stabiliser) certificate, every reflection-preserving automorphism of $\rho(W)$ lies in $\rho(W)\langle\pi_0\rangle$, for all seven types ($\pi_0 = 1$ for D_5, E_6, E_7, E_8).

For each of the seven types the final theorem has exactly four hypotheses, the cited inputs: W is finite; P is characteristic [7]; $P^{ab} \cong \mathbb{Z}^N$ as groups, with $N = |\mathcal{T}|$ (Brieskorn); and the Franzsen–Howlett decomposition $\mathrm{Aut}(W) = R(W)\mathcal{A}(W)$ [22, Thm. 31]. The formalisation uses computer-found constituents rather than the hand constructions of Section 4. Type D_5 is formalised with the same four hypotheses (for D_5 the Franzsen–Howlett input is [22, Thm. 31(3)]); its certificate uses the constituent U^- of Proposition 5.2. The quotient results and Proposition C are not included in this Lean formalisation; the latter has a self-contained calculation and an exact Python checker.

2. The characteristic reflection lattice

Lemma 2.1. *Let $\psi \in \mathrm{Aut}(G)$.*

(a) If $K \trianglelefteq G$ with $\psi(K) = K$, then $R(\psi) \geq R(\bar{\psi})$.
(b) If $G = \mathbb{Z}^m$ and ψ is given by B, then $R(\psi) = |\operatorname{coker}(I - B)|$. This is ∞ if B has eigenvalue 1.
(c) $R(\iota_g \circ \psi) = R(\psi)$, where $\iota_g(x) = gxg^{-1}$.
(d) If $[G:H] < \infty$ and $\psi(H) = H$, then $R(\psi|_H) \leq [G:H]\,R(\psi)$.

Proof. (a) The projection maps classes onto classes.
(b) $y \sim x$ iff $y - x \in (I - B)\mathbb{Z}^m$.
(c) $x \mapsto xg$ maps $\iota_g\psi$-classes bijectively to ψ-classes.

(d) Write $G = \bigsqcup_i Hg_i$. If $y = zx\psi(z)^{-1} \in H$ with $z = hg_i$, then y is $\psi|_H$-conjugate to $g_i x \psi(g_i)^{-1}$. □

Notation. For a Coxeter graph with vertex set S:

- $A = \langle S \mid \Pi(s,t;m_{st}) = \Pi(t,s;m_{st})\ (s \neq t,\ m_{st} < \infty)\rangle$, where $\Pi(s,t;m) = sts\cdots$ has m letters;
- $W = A/\langle\langle s^2 \rangle\rangle$, with quotient map $\mu\colon a \mapsto \bar{a}$, and $P = \ker \mu$;
- $\mathcal{T}$ is the set of reflections, $\rho(w)(r) = wrw^{-1}$, and $\rho(w)$ also denotes the permutation matrix on $\mathbb{Z}^{\mathcal{T}}$.

Since $\mathcal{T}$ generates W, $\ker \rho = Z(W)$. Each $\alpha \in \mathrm{Aut}(W)$ induces θ_α on $\rho(W)$ by $\theta_\alpha \rho(w) = \rho(\alpha w)$. For finite irreducible W of rank ≥ 3, $Z(W) \subseteq \{\pm 1\}$, and $r \mapsto \rho(r)$ is injective on $\mathcal{T}$.

Theorem 2.2 ([7, Thm. 1.4])**.** *If A is irreducible of spherical type, then P is characteristic in A.*

This was extended to reducible types in [21]. The proof in [7] is computer-assisted in several cases:

- for H_4, F_4, E_6, E_7, E_8 it uses GAP calculations [7, Prop. 5.4, Cor. 5.5];
- for H_3 it uses one GAP calculation [7, Lemma 5.1];
- among the classical types it uses GAP for B_4, B_6, D_4, D_6 [7, Lemmas 3.5, 4.4].

We have not re-run these calculations. Each $\varphi \in \mathrm{Aut}(A)$ thus induces $\varphi_W \in \mathrm{Aut}(W)$ and $\theta_\varphi := \theta_{\varphi_W}$.

Lemma 2.3. *For any Coxeter group, $s \mapsto (e_s, s)$ defines a homomorphism $f\colon A \to \mathbb{Z}^{\mathcal{T}} \rtimes W$, with $(v,w)(v',w') = (v + w\cdot v', ww')$ and $w \cdot e_r = e_{wrw^{-1}}$.*

Proof. For $\Pi(s,t;m)$, the $\mathbb{Z}^{\mathcal{T}}$-component is $\sum_{k=1}^{m} e_{r_k}$ with $r_k = (st)^{k-1}s$. For $\Pi(t,s;m)$ it is $\sum_k e_{r'_k}$ with $r'_k = (st)^{m-k}s$. Both sets equal $\{(st)^j s : 0 \leq j < m\}$. Compare [7, Lemma 3.7]. □

Proposition 2.4. *Let A be of spherical type. Then $f(P) = 2\mathbb{Z}^{\mathcal{T}}$, and f induces an isomorphism $H_1(P;\mathbb{Z}) \cong 2\mathbb{Z}^{\mathcal{T}}$ under which conjugation by a acts as the permutation $\rho(\bar{a})$.*

Proof. We have $f(as^2a^{-1}) = (2e_{\bar{a}s\bar{a}^{-1}}, 1)$, and these elements generate $f(P)$. Moreover $P \cong \pi_1$ of the complexified reflection-arrangement complement [2], whose H_1 is free on the meridians, one per hyperplane [3, 32]. So $H_1(P;\mathbb{Z})$ has rank $|\mathcal{T}|$, and the surjection onto $2\mathbb{Z}^{\mathcal{T}}$ is an isomorphism. □

Corollary 2.5. *Let M_φ be the matrix of φ_* on $\mathbb{Z}^{\mathcal{T}}$. Then $M_\varphi \rho(w) M_\varphi^{-1} = \theta_\varphi(\rho(w))$ and $M_{\iota_a \circ \varphi} = \rho(\bar{a}) M_\varphi$. Hence θ_φ preserves the permutation character $\chi_{\mathcal{T}}(g) = \#\{r : g(r) = r\}$.*

The image $f(A) \cong A/[P,P]$ is the crystallographic quotient of [23, 31]; see also [14].

3. The fixed-vector criterion

A $\rho(W)$-invariant subspace $U \subseteq \mathbb{Q}^{\mathcal{T}}$ is *good* if $U \otimes \mathbb{C}$ is irreducible and occurs with multiplicity one in $\mathbb{C}^{\mathcal{T}}$. A good U is a full isotypic component.

Lemma 3.1. *If U is good and $N \in GL(\mathbb{Z}^{\mathcal{T}})$ commutes with $\rho(W)$, then $NU = U$ and $N|_U = \pm 1$.*

Proof. $NU \cong U$, so $NU = U$. By Schur's lemma $N|_U = c \in \mathbb{Q}^\times$. Applying $N^{\pm 1}$ to a primitive vector of $U \cap \mathbb{Z}^{\mathcal{T}}$ gives $c^{\pm 1} \in \mathbb{Z}$. □

Lemma 3.2. *A nonzero invariant $U \subseteq \mathbb{Q}^{\mathcal{T}}$ is good in each of the following cases.*

(i) Every orbital matrix of $\rho(W)$ on $\mathcal{T} \times \mathcal{T}$ acts on U by a scalar.
(ii) $\langle \chi_U, \chi_U \rangle = \langle \chi_U, \chi_{\mathcal{T}} \rangle = 1$.
(iii) $U \otimes \mathbb{C}$ is irreducible and $\sum_r \dim(U \otimes \mathbb{C})^{\mathrm{Stab}(r)} = 1$, where r runs over representatives of the $\rho(W)$-orbits on $\mathcal{T}$ (Frobenius reciprocity).

Proof. (i) The orbital matrices span the commutant $\mathcal{C} = \mathrm{End}_{\mathbb{C}W}(\mathbb{C}^{\mathcal{T}}) = \prod_i M_{m_i}(\mathbb{C})$, where $\mathbb{C}^{\mathcal{T}} = \bigoplus_i V_i \otimes \mathbb{C}^{m_i}$. By hypothesis $\mathcal{C}$ maps $U_{\mathbb{C}} = U \otimes \mathbb{C}$ into itself and acts on it by scalars. The central idempotents $e_i \in \mathcal{C}$ act on $U_{\mathbb{C}}$ by 0 or 1 and sum to 1, so exactly one e_i acts as 1. Hence $U_{\mathbb{C}} \subseteq V_i \otimes \mathbb{C}^{m_i}$ meets only one isotypic component. Being W-invariant, $U_{\mathbb{C}} = V_i \otimes L$ for some $0 \neq L \subseteq \mathbb{C}^{m_i}$, and $M_{m_i}(\mathbb{C})$ preserves L and acts on it by scalars. This forces $m_i = 1$ and $L = \mathbb{C}$. (ii) and (iii) are standard. □

Theorem 3.3. *Let A be irreducible of spherical type. Suppose that for every $\varphi \in \mathrm{Aut}(A)$ there is a permutation π of $\mathcal{T}$ with $\pi\rho(w)\pi^{-1} = \theta_\varphi(\rho(w))$ for all w, such that one of the following holds.*

(i) (stable) *There is a good U with $\pi U = U$ such that, for each $\varepsilon = \pm 1$, some $h\pi$ ($h \in \rho(W)$) has eigenvalue ε on U.*

(ii) (swap) *There are good $U_1 \neq U_2$ with $\pi U_1 = U_2$ such that, for each $\varepsilon = \pm 1$, some $(h\pi)^2$ has an eigenvector in U_1 with eigenvalue ε.*

Then A has property R_∞.

Proof. $N := \pi^{-1} M_\varphi$ commutes with $\rho(W)$.

Case (i). $N|_U = c = \pm 1$. Choose $h = \rho(\bar{a})$ and $0 \neq v \in U$ with $h\pi v = cv$, and let $\varphi' = \iota_a \circ \varphi$. Then $M_{\varphi'} v = h\pi N v = v$.

Case (ii). $N|_{U_i} = c_i$. Choose $h = \rho(\bar{a})$ and $0 \neq v \in U_1$ with $(h\pi)^2 v = c_1 c_2 v$. With $Q = h\pi$ we have $QU_1 = U_2$, and $M^2_{\varphi'} v = QNQNv = v$. Since $M_{\varphi'} v \in U_2$ and $U_1 \cap U_2 = 0$, $u = v + M_{\varphi'} v$ is a nonzero fixed vector.

So $M_{\varphi'}$ fixes a nonzero vector of $\mathbb{Z}^{\mathcal{T}}$. By Lemma 2.1(b),(a) and Proposition 2.4, $R(\varphi'|_P) = \infty$. Then $R(\varphi') = \infty$ by (d), and $R(\varphi) = R(\varphi')$ by (c). □

Remark 3.4 (Relation to the crystallographic criteria)**.** By Marin [31, Thm. 2.1], the quotient $\Gamma = A/[P,P] \cong f(A)$ is crystallographic with holonomy group $W/Z(W)$. Its translation lattice is the preimage of $Z(W)$, which contains $P/[P,P]$ with finite index. (Marin also shows that the holonomy is W itself iff $Z(W) = 1$ [31, Cor. 2.2].) In this language, Theorem 3.3 is close to the Dekimpe–Penninckx criterion [12, Cor. 3.10], [11, Thm. 3.3] for crystallographic groups: $R = \infty$ iff $\det(1 - AD) = 0$ for some holonomy element A. It is combined with the multiplicity-one device of [8, Thm. 4.7] to produce the eigenvector. We work directly with P so that the reduction stays in terms of the characteristic pure subgroup and the induced automorphisms of W. See Theorem 6.2 for the consequence for Γ itself.

Remark 3.5.

- If π works for θ, then $g\pi$ works for $\iota_g\theta$, since $\{hg\pi\} = \{h\pi\}$. So one θ per coset of $\mathrm{Inn}(\rho(W))$ suffices.
- For inner θ take $\pi = 1$. Then (i) holds for any good U on which some reflection acts nontrivially. This is automatic for a nontrivial good U when all reflections are conjugate: the kernel of the action is normal, so if it contained one reflection it would contain all of them.
- If $r \mapsto \rho(r)$ is injective on $\mathcal{T}$ (true in rank ≥ 3, and for odd dihedral types) and $\theta(\rho(\mathcal{T})) = \rho(\mathcal{T})$, then $\pi(r) = \rho^{-1}\theta\rho(r)$ realises θ.

4. The exceptional types

4.1. **Induced automorphisms.** For finite irreducible W, write $R(W)$ for its reflection-preserving automorphism group and $\mathrm{Gr}(W)$ for its graph automorphism group. A *central twist* $w \mapsto w\,c(w)$, with $c\colon W \to Z(W)$ a homomorphism, induces the identity on $W/Z(W)$. Franzsen and Howlett prove the following [22, Thm. 31, pp. 329–330].

- For W of type $B_n, D_n, E_6, E_7, E_8, F_4, H_3, H_4$, $\mathrm{Aut}(W) = R(W)\mathcal{A}(W)$, where $\mathcal{A}(W)$ consists of central twists.
- $R(W) = \mathrm{Inn}(W)\,\mathrm{Gr}(W)$ except for H_3 and H_4, where $[R(W) : \mathrm{Inn}(W)] = 2$.

In [22] (in the published version as in the preprint) the proof of Theorem 31 is given as an outline, based on the sizes of the conjugacy classes of involutions. (In item (9) of that theorem, for F_4, "α_χ" should read α_ζ.) For the six exceptional types, Appendix B gives independent control of all of $\mathrm{Aut}(\rho(W))$: by exhaustive computation for E_6, E_7, F_4, H_3, H_4, and by hand for E_8.

Lemma 4.1. *Let $\varphi \in \mathrm{Aut}(A)$, with A of type X. Then $\theta_\varphi \in \mathrm{Inn}(\rho(W))\,\Theta_X$, where:*

- $\Theta_X = \{1\}$ *for* $X = E_6, E_7, E_8, B_n$ *($n \geq 3$);*
- $\Theta_{F_4} = \{1, \theta_L\}$, $\Theta_{H_3} = \{1, \theta_\Sigma\}$, $\Theta_{H_4} = \{1, \theta_\Psi\}$;
- $\Theta_{D_n} = \{\theta_\gamma : \gamma \in \mathrm{Gr}(D_n)\}$.

Every element of Θ_X maps $\rho(\mathcal{T})$ onto itself, so it is realised by a permutation of $\mathcal{T}$.

Proof. Central twists act trivially on $\rho(W)$, so θ_φ is induced by an element of $R(W)$.

- For E_6 (and D_n, n odd) the graph automorphism is induced by conjugation by w_0, hence inner. For E_7, E_8 and B_n, Gr is trivial.
- For F_4, conjugation by the similarity $L(x) = (x_1 + x_2, x_1 - x_2, x_3 + x_4, x_3 - x_4)$ satisfies $Ls_\alpha L^{-1} = s_{L\alpha}$. It is reflection-preserving and exchanges the two reflection classes, so it is not inner. Since $[R(W) : \mathrm{Inn}(W)] \leq |\,\mathrm{Gr}(W)| = 2$, it represents the nontrivial coset. It remains non-inner on $\rho(W)$: equality of two reflection images there forces equality of the reflections, since $-s$ has a 3-dimensional (-1)-eigenspace and is not a reflection in rank four.
- For H_3, resp. H_4, conjugation by Σ, resp. Ψ (Propositions 4.3 and 4.6) is reflection-preserving and induces a non-inner automorphism of $\rho(W)$. It therefore represents the nontrivial coset of $\mathrm{Inn}(W)$ in $R(W)$.

Finally, $\theta_r(\rho(\mathcal{T})) = \rho(\mathcal{T})$ for $r \in R(W)$, by definition of $R(W)$. □

4.2. **Explicit good constituents.**

Proposition 4.2 (E_6, E_7, E_8)**.** *Let $A_\perp$ be the adjacency matrix of the orthogonality graph on root lines.*

(a) $\rho(W)$ has exactly three orbitals on $\mathcal{T} \times \mathcal{T}$: equality, orthogonality and non-orthogonality.
(b) The graph is strongly regular with parameters $(36, 15, 6, 6)$, $(63, 30, 13, 15)$, $(120, 63, 30, 36)$. *Its nontrivial eigenvalues are* $3^{15}, (-3)^{20}$; $3^{35}, (-5)^{27}$; $3^{84}, (-9)^{35}$ *(multiplicities as exponents).*
(c) The two nontrivial eigenspaces are good, and every reflection acts nontrivially on them.

Proof. By Steinberg's theorem [38], the stabiliser of a root, or of a pair of roots, is the Weyl group of the perpendicular root subsystem. The perpendicular subsystems are listed below. The root counts and ranks are finite computations, certified exactly (exact rational ranks) as proof certificates in Appendix A. Each pair (number of roots, rank) in the table occurs for exactly one simply-laced root system, irreducible or not, of that rank, so the counts determine the types. These are the classical facts that the centraliser of a reflection in $W(E_8)$, $W(E_7)$, $W(E_6)$ involves $W(E_7)$, $W(D_6)$, $W(A_5)$.

	$\alpha^\perp$	$\{\alpha, \gamma\}^\perp$, $\alpha \perp \gamma$	$\{\alpha, \beta\}^\perp$, $(\alpha, \beta) = 1$
E_6	A_5 (30, rank 5)	A_3 (12, rank 3)	A_2^2 (12, rank 4)
E_7	D_6 (60, rank 6)	D_4A_1 (26, rank 5)	A_5 (30, rank 5)
E_8	E_7 (126, rank 7)	D_6 (60, rank 6)	E_6 (72, rank 6)

W is transitive on roots. Inside $W(\alpha^\perp)$, the orbits are as follows.

- The orbit of an orthogonal root has size $720/24 = 30$, $23040/384 = 60$, $2903040/23040 = 126$, i.e. all of $\alpha^\perp$.
- For $(\alpha, \beta) = 1$, the orbit sizes are $720/36 = 20$, $23040/720 = 32$ and $2903040/51840 = 56$. Each orbit consists of all such roots.

Passing to lines gives (a). Then k, λ, μ are half the root counts in the table, and (b) follows from the standard formulas for strongly regular graphs.

By (a) the commutant is spanned by I, $A_\perp$ and J. These act by scalars on the nontrivial eigenspaces, which are rational because the eigenvalues are integers. So Lemma 3.2(i) applies, and Remark 3.5 gives the last assertion. □

Proposition 4.3 (H_3)**.** *(a) $\rho(W) \cong A_5$. The 15 root lines form 5 orthogonal triples (*frames*), which correspond to the Klein four-subgroups.*

(b) The frame module U (functions constant on frames, with sum 0) is a good 4-dimensional constituent. More precisely, writing irreducible A_5-modules by their dimensions,

$$\mathbb{Q}^{\mathcal{T}} \cong 1 \oplus U \oplus 5 \oplus 5.$$

(c) For $\Sigma = \varsigma \circ \sigma$, where σ is Galois conjugation $\sqrt{5} \mapsto -\sqrt{5}$ on coordinates and ς swaps the last two coordinates, $\Sigma\Phi = \Phi$ and $\Sigma s_\alpha \Sigma^{-1} = s_{\Sigma\alpha}$. The induced π_Σ is not inner on $\rho(W)$.

(d) For every outer θ there is h such that $h\pi$ acts on the frames as a transposition. It then has eigenvalues $+1$ (multiplicity 3) and -1 (multiplicity 1) on U.

Proof. (a) Commuting reflections correspond to orthogonal lines, and the 15 involutions of A_5 lie in 5 Klein four-subgroups, permuted as A_5 permutes 5 points.

(b) U is the augmentation module of a 2-transitive action, hence absolutely irreducible. Since $\mathbb{Q}^{\mathcal{T}} = \mathrm{Ind}_{V_4}^{A_5} 1$, the multiplicity of an irreducible χ is $\dim \chi^{V_4}$. This equals $1, 0, 0, 1, 2$ for χ of degree $1, 3, 3', 4, 5$, which gives the displayed decomposition.

(c) Σ satisfies $(\Sigma x, \Sigma y) = \sigma(x, y)$. θ_Σ is not inner: on the rotation subgroup $W^+ \cong \rho(W)$ it Galois-conjugates the character of $\mathbb{R}^3$ on elements of order 5, while inner automorphisms preserve characters. This is also checked by machine.

(d) An outer automorphism of A_5 is conjugation by an odd permutation, and π permutes the frames accordingly. So $\{h\pi\}$ contains a transposition. In fact π_Σ itself is a transposition on frames. □

Proposition 4.4 (F_4)**.** *Each class of 12 root lines (long, short) splits into 3 orthogonal frames of 4 lines. Let $U_{\mathrm{long}}, U_{\mathrm{short}}$ be the corresponding frame modules (sum zero). Then:*

(a) both are good and 2-dimensional;

(b) long reflections act on U_{long} as transpositions of frames and trivially on U_{short}, and vice versa;

(c) π_L exchanges U_{long} and U_{short}, and $\pi_L^2 = 1$;

(d) for a long simple reflection s, $(s\pi_L)^2 = s \cdot \pi_L s \pi_L$ has eigenvalue -1 on U_{short}, and π_L^2 gives eigenvalue $+1$.

Hence case (ii) of Theorem 3.3 holds for θ_L.

Proof. The long roots $\pm e_i \pm e_j$ form a D_4, whose frames correspond to the pairings $ij|kl$. The short lines e_i form one frame. The lines $\frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1)$ form two frames, according to the parity of the number of minus signs. Each frame system is permuted through all of S_3, so each frame module is the standard module of S_3.

Multiplicity one (Lemma 3.2(iii)):

- the stabiliser of the long line $e_1 + e_2$ fixes one long frame and contains $s_{e_1 - e_2}$, which swaps the other two, so $\dim U_{\mathrm{long}}^{\mathrm{Stab}} = 1$;
- the stabiliser of the short line e_1 contains $\mathrm{Sym}\{2, 3, 4\}$, which induces S_3 on pairings, so $\dim U_{\mathrm{long}}^{\mathrm{Stab}} = 0$;
- U_{short} follows by applying π_L.

(b) Check on $s_{e_1 - e_2}$ and s_{e_1}, and use conjugacy.

(c) L maps orthogonal frames to orthogonal frames and long to short lines, and $L^2 = 2I$.

(d) $\pi_L s \pi_L$ is a short reflection. □

Lemma 4.5 (Icosian model of H_4)**.** *Halve the* 120 *roots of* H_4 *(Appendix A) and read the first coordinate as the real part. These form a group* I *of unit quaternions (the binary icosahedral group), and* $K = I/\{\pm 1\} \cong A_5$*. Identify* $\mathcal{T}$ *with* K*. Here* $K \cong A_5$ *is certified by a finite computation (Appendix A): K has* 60 *elements and* 15 *involutions, which lie in* 5 *Klein four-subgroups; conjugation on these gives an injective homomorphism* $K \to S_5$ *whose image consists of even permutations. Then:*

(a) the reflection in a unit root r *is* $x \mapsto -r\bar{x}r$*, and it acts on* K *by* $x \mapsto rx^{-1}r$*;*
(b) $\rho(W) = \{x \mapsto ax^{\pm 1}b : a, b \in K\} \cong (K \times K) \rtimes C_2$*.*

Proof. (a) For $|r| = 1$, $r\bar{x}r = 2(x, r)r - x$, using $\bar{x}r + \bar{r}x = 2(x, r)$.

(b) By (a), $\rho(W)$ is contained in the right-hand group, which has order $7200 = |W|/2 = |\rho(W)|$. The closure of I under multiplication is a finite computation, certified exactly in Appendix A. □

Proposition 4.6 (H_4)**.** *Let* V *be the* 4*-dimensional augmentation module of* $K \cong A_5$ *acting on its five Klein four-subgroups, and let* $U \subseteq \mathbb{Q}^K$ *be the span of the matrix coefficients* $x \mapsto \mathrm{tr}(X\lambda_V(x))$*, so* $U \cong \mathrm{End}(V)$ *has dimension* 16*.*

(a) U *is good.*
(b) Let $\Psi = \Omega \circ \sigma$*, where* σ *is Galois conjugation and* Ω *is the rotation* $j \mapsto k$*,* $k \mapsto -j$*. Then* Ψ *is a ring automorphism of* $\mathbb{H}_{\mathbb{Q}(\sqrt{5})}$ *with* $\Psi(I) = I$*. The induced permutation* π_Ψ *normalises* $\rho(W)$*, preserves reflections, and is not inner.*
(c) For every outer θ_φ *induced by* $\mathrm{Aut}(A)$ *(and in fact every outer automorphism of* $\rho(W)$*, by Appendix B) there is* h *such that* $h\pi$ *acts on* U *as conjugation by a transposition. This has eigenvalues* $+1$ *(multiplicity* 10*) and* -1 *(multiplicity* 6*).*

Proof. (a) By Wedderburn, $\mathbb{C}[K] = \bigoplus_{V'} V' \otimes V'^*$ as a $K \times K$-module, each summand irreducible and occurring once. The summand for $V_\mathbb{C}$ is $U \otimes \mathbb{C} \cong V_\mathbb{C} \otimes V_\mathbb{C}^*$, which is preserved by inversion. Since V is defined over $\mathbb{Q}$, so is $U \cong \mathrm{End}_\mathbb{Q}(V)$. Hence U is good for $\rho(W)$. (We do not need the remaining summands. Over $\mathbb{C}$ they are 1, $3 \otimes 3$, $3' \otimes 3'$ and $5 \otimes 5$; over $\mathbb{Q}$ the two 9-dimensional ones combine into one 18-dimensional summand.)

(b) Ψ commutes with quaternion conjugation, so $\Psi s_r \Psi^{-1} = s_{\Psi r}$. The equality $\Psi(I) = I$ is a finite computation, certified exactly in Appendix A. If θ_Ψ were inner, then $\pi_\Psi \in \rho(W)$, because the centraliser of $\rho(W)$ in $\mathrm{Sym}(K)$ is trivial. Indeed, a permutation commuting with all left and right translations is a translation by an element of $Z(K) = 1$. Since π_Ψ fixes 1, it would be $x \mapsto ax^{\pm 1}a^{-1}$. The sign $-$ is impossible, since π_Ψ is an automorphism of the non-abelian group K. The sign $+$ is impossible, since conjugation preserves real parts up to sign, whereas Ψ maps a quaternion with real part $\tau/2$ (it has order 10 in I and represents an element of order five in K) to one with real part $-\tau^{-1}/2$ (here $\tau = \frac{1+\sqrt{5}}{2}$).

(c) We argue in three steps.

(i) By Lemma 4.1 and Remark 3.5 (or by Appendix B for an arbitrary outer θ), we may take $\pi \in \rho(W)\pi_\Psi$. So $\{h\pi : h \in \rho(W)\} = \{h\pi_\Psi\}$.
(ii) By (b), $\pi_\Psi = \psi$ is an outer automorphism of K.
(iii) Since $\mathrm{Aut}(A_5) \cong S_5$, acting on the five Klein four-subgroups, ψ is conjugation by an odd permutation δ'. For $a \in K = A_5$, the element $h = (x \mapsto axa^{-1})$ lies in $\rho(W)$, and $h\psi$ is conjugation by $a\delta'$. Choose a with $a\delta' = \delta$ a transposition.

The action of K on V extends to S_5, with $\lambda_V(\delta)$ having eigenvalues $1, 1, 1, -1$. Since $h\pi_\Psi$ acts on $U \cong \mathrm{End}(V)$ as $X \mapsto \lambda_V(\delta)X\lambda_V(\delta)^{-1}$, the eigenvalue $+1$ has multiplicity $9 + 1$ and -1 has multiplicity 6. □

Proof of the new cases in Theorem A. By Lemma 4.1 and Remark 3.5, it suffices to treat Θ_X.

- E_n: only the inner class occurs. Proposition 4.2 supplies U, and Remark 3.5 applies, since all reflections are conjugate.

- H_3, H_4: the inner class is covered by Remark 3.5 (all reflections are conjugate and U is nontrivial). The outer class is covered by Propositions 4.3 and 4.6: there $\pi U = U$, since U is the only good constituent of its dimension, and case (i) applies.
- F_4: there are two classes of reflections, so the inner class is treated explicitly. Take $\pi = 1$ and $U = U_{\mathrm{long}}$; then $h = 1$ gives $+1$, and a long reflection gives -1 by Proposition 4.4(b), so case (i) holds. The outer class θ_L is covered by Proposition 4.4, case (ii).
- D_5: Proposition 5.2.

Apply Theorem 3.3. □

type	$\lvert\mathcal{T}\rvert$	$\lvert\rho(W)\rvert$	Θ_X	good constituent used (dim)	outer witness
E_6	36	51840	1	**SRG** eigenspace (15; also 20)	—
E_7	63	1451520	1	**SRG** eigenspace (35; also 27)	—
E_8	120	348364800	1	**SRG** eigenspace (84; also 35)	—
F_4	24	576	$\{1, \theta_L\}$	$\mathbf{U_{long}}, \mathbf{U_{short}}$ (2+2)	swap: π_L^2, $(s\pi_L)^2$
H_3	15	60	$\{1, \theta_\Sigma\}$	**frame** module (4)	eigen-multiplicities 3, 1
H_4	60	7200	$\{1, \theta_\Psi\}$	End(**V**) (16)	eigen-multiplicities 10, 6

TABLE 1. The exceptional-type data used for Theorems A and B. The dimensions agree with independent computer certificates (Appendix A).

5. Uniform proofs for the classical types

Except for D_5, these results are known [5, 35]. We include them to illustrate the method, and because it gives a uniform treatment of D_5 (Remark 5.3).

Proposition 5.1 (A_n, $n \geq 2$)**.** *$A(A_n)$ has property R_∞.*

Proof. Here $\rho(W) = S_{n+1}$ acts on the transpositions, i.e. on the 2-subsets. By Young's rule $\mathbb{Q}^{\mathcal{T}} = S^{(n+1)} \oplus S^{(n,1)} \oplus S^{(n-1,2)}$ (for $n = 2$ the last summand is absent). So $S^{(n,1)}$ is good. For $n+1 \neq 6$, $\mathrm{Aut}(S_{n+1}) = \mathrm{Inn}$. For $n+1 = 6$, an outer automorphism maps $(12) \mapsto (12)(34)(56)$, but $\chi_{\mathcal{T}}$ takes the values 7 and 3 there, so it is not of the form θ_φ (Corollary 2.5). Apply Remark 3.5. □

For B_n and D_n let $u_{ij} = e_{[e_i+e_j]} - e_{[e_i-e_j]}$ $(i < j)$ and $U^- = \mathrm{span}\{u_{ij}\}$. Let $E \leq W$ be the group of sign changes: all of them for B_n, the even ones for D_n.

Proposition 5.2 (B_n, $n \geq 3$; D_n, $n \geq 5$)**.** *U^- is good, and $A(B_n)$ and $A(D_n)$ have property R_∞.*

Proof. Weights. E acts on u_{ij} by $\chi_{ij}(\varepsilon) = \varepsilon_i\varepsilon_j$. These characters are nontrivial and pairwise distinct on E. For D_n this needs $n \geq 5$: on even sign changes $\chi_{ij} = \chi_{kl}$ iff $\varepsilon_i\varepsilon_j\varepsilon_k\varepsilon_l$ is trivial or the product of all signs.

Goodness. E fixes $e_{[e_i+e_j]} + e_{[e_i-e_j]}$ and the short lines. So each χ_{ij} occurs once in $\mathbb{Q}^{\mathcal{T}}$. Since $S_n \leq W$ permutes the weights transitively, $U^- \otimes \mathbb{C}$ is irreducible.

Automorphisms. By Lemma 4.1, θ_φ is inner for B_n, and inner up to the diagram involution for D_n (for odd n the latter is inner too). For the inner class, $\pi = 1$ works: the reflection s_{e_n} (B_n), resp. $s_{e_{n-1}-e_n}$ (D_n), acts nontrivially on U^-. The diagram involution is realised by negating e_n, which acts on u_{ij} by -1 if $n \in \{i, j\}$ and by $+1$ otherwise. So case (i) holds with $h = 1$. □

Remark 5.3 (Type D_5)**.** Proposition 5.2 with $n = 5$ proves R_∞ for $A(D_5)$. Its inputs are [7], [22] (here $\mathrm{Aut}(W(D_5)) = R(W) \cong W$ by [22, Thm. 31(3)]) and the weight argument.

There is also a short derivation from Paris–Soroko's description $\mathrm{Aut}(A[D_5]) = \mathrm{Inn}(A) \rtimes \langle\chi\rangle$, with χ the inversion of the generators [33, Thm. 1]:

- $R(\iota_g) = R(\mathrm{id}) = \infty$ by Lemma 2.1(c), since $A^{ab} \cong \mathbb{Z}$ and Lemma 2.1(a),(b) give infinitely many conjugacy classes;
- $R(\iota_g \circ \chi) = R(\chi) = \infty$ by Remark 5.6(b).

This deduction is nearly immediate once [33, Thm. 1] is available, but we have not found it written down. [35] states D_5 as a conjecture, and [33] does not mention R_∞. We therefore record D_5 as part of Theorem A.

Proposition 5.4 (D_4)**.** *In type D_4:*

- *U^- is reducible: $J(u_{ij}) = u_{kl}$ ($\{k,l\}$ complementary) is a non-scalar operator on U^- commuting with $W(D_4)$;*
- *the frame module U on the 3 orthogonal frames $\{12|34, 13|24, 14|23\}$ is good of dimension 2;*
- *$A(D_4)$ has property R_∞.*

Proof. For J: a permutation $\sigma \in S_4$ maps u_{ij} to $u_{\sigma i\, \sigma j}$ and commutes with taking complements; an even sign change ε acts by $\varepsilon_i\varepsilon_j$ on u_{ij}, and $\varepsilon_i\varepsilon_j = \varepsilon_k\varepsilon_l$ because $\varepsilon_1\varepsilon_2\varepsilon_3\varepsilon_4 = 1$. Hence J commutes with $W(D_4) = E \rtimes S_4$, and its ±1-eigenspaces are proper invariant subspaces of U^-. The E-fixed part of $\mathbb{Q}^{\mathcal{T}}$ is the permutation module on the 6 pairs, namely $1+3+2$ for S_4. Its 2-dimensional summand is U, which does not occur in U^-. Every θ_φ preserves reflections, so π permutes the frames. W induces all of S_3 on frames, so some $h\pi$ is a transposition there, with eigenvalues ±1 on U. □

Remark 5.5 (Dihedral types)**.** Let m be odd. Then $Z(W) = 1$, so ρ is faithful and injective on $\mathcal{T}$, and $\mathbb{Q}^{\mathcal{T}} \cong \bigoplus_{d|m} U_d$ with $\dim U_d = |(\mathbb{Z}/d\mathbb{Z})^\times|$, and U_d is absolutely irreducible iff $d \in \{1, 3\}$.

- If $3 \nmid m$ there is no nontrivial good constituent, and Theorem 3.3 cannot be applied. For m prime, Schur's lemma gives only a unit of $\mathbb{Q}(\cos\frac{2\pi}{m})$.
- If $3 \mid m$, then U_3 is the unique good constituent of dimension 2, so $\pi U_3 = U_3$. The group $\rho(W)$ acts on U_3 as $S_3 \le O(2)$. Since π is orthogonal and normalises S_3, it acts as an element of $S_3 \times \{\pm1\}$. Hence some $h\pi$ acts as $\pm$(a reflection), with eigenvalues ±1. So Theorem 3.3 applies, using [7] and that every automorphism of W preserves reflections.

This was confirmed by computation for odd $m \le 21$. In any case R_∞ for $I_2(m)$ is known [5]; Theorem 6.6 below treats the quotients for all m. For $m = 5$ the failure of this criterion reflects an actual obstruction at the quotient level: Proposition 6.7 proves that $A(I_2(5))/[P,P]$ does not have R_∞.

Remark 5.6 (Two elementary observations)**.**

(a) If φ fixes the generator $z \in \{\Delta, \Delta^2\}$ of $Z(A)$ [4, 13], then φ_* fixes the nonzero image of z in $H_1(A)$, so $R(\varphi) = \infty$. Only centre-inverting automorphisms need the criterion.

(b) For any Artin group that is not free abelian, the inversion $s \mapsto s^{-1}$ has $R = \infty$. Choose $m_{st} \ge 3$. Via f, $\max_r |v_r - v_{srs}|$ is an invariant of twisted classes with W-part s, and it separates the elements $t^{2k}s$. Cf. [34, Thm. 1], [27].

6. Crystallographic quotients

6.1. **A criterion on the translation lattice.** Let L be a lattice, let $V = L \otimes \mathbb{Q}$, and let $G \le GL(L)$ be finite. Call a rational G-submodule of V *good* if its complexification is irreducible of multiplicity one in $V_{\mathbb{C}}$.

Proposition 6.1. *Let Γ be crystallographic with translation lattice L and holonomy G. Suppose that every linear part D of an automorphism of Γ admits $\pi \in GL(L)$ inducing the same automorphism of G, such that either the stable or the swap hypothesis of Theorem 3.3 holds with G, V in place of $\rho(W), \mathbb{Q}^{\mathcal{T}}$. Then Γ has property R_∞.*

Proof. The operator $N = \pi^{-1}D$ centralises G. On each good constituent it is a rational scalar; preservation of $U \cap L$ by N and N^{-1} forces this scalar to be ± 1. The stable and swap arguments in Theorem 3.3 give $g \in G$ such that gD has a nonzero fixed vector. Lift g to $\gamma \in \Gamma$ and apply Lemma 2.1 to $\iota_\gamma \Phi$ and its restriction to the characteristic finite-index subgroup L. □

The stable case is a lattice reformulation of the multiplicity-one device in [8, Thm. 4.7], using the crystallographic criterion [12, Cor. 3.10] to allow torsion. The swap case also covers exchanged constituents, which do not satisfy the non-conjugacy hypothesis of that stable criterion. Only automorphisms induced by the actual extension need be considered. More precisely, if Θ_Γ denotes their image in $\mathrm{Aut}(G)$, then

$$\Theta_\Gamma \subseteq \Theta_\chi := \{\theta \in \mathrm{Aut}(G) : \chi_V \circ \theta = \chi_V\},$$

because their linear parts intertwine the holonomy representations. Controlling all of $\mathrm{Aut}(G)$ is a sufficient, sometimes convenient, stronger condition.

6.2. **The seven positive cases.**

Theorem 6.2 (Seven positive quotients)**.** *For $X \in \{D_5, E_6, E_7, E_8, F_4, H_3, H_4\}$, the crystallographic group $\Gamma = A(X)/[P, P]$ has property R_∞.*

Proof. Identify Γ with $f(A) \leq \mathbb{Z}^{\mathcal{T}} \rtimes W$, acting on $\mathbb{R}^{\mathcal{T}}$ by $(v, w) \cdot x = v + \rho(w)x$.

The translation lattice. Let $\Lambda = \{(v, w) \in \Gamma : w \in Z(W)\}$, the preimage of $Z(W)$ [31, Thm. 2.1]. Its elements act as translations by v. This action is faithful: elements with W-part 1 have even coordinates, while those with W-part w_0 (if $w_0 = -1$) have odd coordinates, since $f(\Delta) = (\sum_r e_r, w_0)$. So $\Lambda \cong L$, where $L = 2\mathbb{Z}^{\mathcal{T}}$ if $Z(W) = 1$ and $L = 2\mathbb{Z}^{\mathcal{T}} + \mathbb{Z}\mathbf{1}$ otherwise. Every permutation matrix preserves L. Since $\ker \rho = Z(W)$, the resulting affine action is faithful. Its full translation lattice has finite index, so the action is discrete and cocompact. This description also applies to the classical and dihedral types below.

Λ *is characteristic.* $\Gamma/\Lambda = \rho(W)$ acts faithfully on L. Let N' be an abelian normal subgroup of finite index. For $n \in N'$, the linear part of n fixes the finite-index sublattice $N' \cap \Lambda$ pointwise, so it is trivial and $n \in \Lambda$. Thus Λ is the unique maximal abelian normal subgroup of finite index, hence characteristic.

The argument. Let $\Phi \in \mathrm{Aut}(\Gamma)$. It preserves Λ and induces $\theta \in \mathrm{Aut}(\rho(W))$ and $D = \Phi|_\Lambda \in GL(L)$ with $D\rho(g)D^{-1} = \theta(\rho(g))$. By Appendix B (for E_8, Proposition B.2; for the other types, the exhaustive computations), θ is realised by a permutation $\pi \in \rho(W)\langle\pi_0\rangle$, and π preserves L. Then $N = \pi^{-1}D \in GL(L)$ commutes with $\rho(W)$. On a good constituent U, N acts as a rational scalar preserving the lattice $U \cap L$, so as ± 1.

The witnesses of Sections 4 and 5 depend only on $\rho(W)$, U and π. They give $\gamma \in \Gamma$ such that $\Phi' = \iota_\gamma \circ \Phi$ has $\Phi'|_\Lambda$ with a nonzero fixed vector. Hence $R(\Phi'|_\Lambda) = \infty$ by Lemma 2.1(b), so $R(\Phi') = \infty$ by Lemma 2.1(d), and $R(\Phi) = R(\Phi')$ by Lemma 2.1(c). □

The proof uses all automorphisms of $\rho(W)$, not only those induced from $\mathrm{Aut}(W)$. The calculations in Appendix B are therefore necessary inputs to this explicit proof.

6.3. **The classical families.** The distinction between all holonomy automorphisms and those preserving the reflection character is useful in the classical families. Write $\chi_{\mathcal{T}}$ for the permutation character on root lines.

Lemma 6.3. *For W of type B_n, $n \geq 3$, or D_n, $n \geq 5$, every automorphism of $W/Z(W)$ preserving $\chi_{\mathcal{T}}$ preserves the images of reflections. In type B_n it preserves the short and long classes separately.*

Proof. An involution in $W/Z(W)$ lifts to a signed permutation w with $w^2 = I$ or $w^2 = -I$. In the second case w fixes no root line, since a real invariant line would give a real eigenvalue whose square is -1. Thus its character value is zero, unlike that of a reflection in the indicated ranks.

For $w^2 = I$, conjugate the signed transposed pairs to positive pairs and write (a, b, c) for the numbers of positive fixed coordinates, negative fixed coordinates, and transposed pairs; $a+b+2c = n$. Counting fixed root lines gives

$$\chi_D(w) = a(a-1) + b(b-1) + 2c, \qquad \chi_B(w) = a^2 + b^2 + 2c.$$

These counts do not depend on the signs on a transposed pair. In type D_n we require b even.

For a long reflection of B_n, the value is $(n-2)^2 + 2$. Equality is equivalent to

$$ab = 2n(1-c) + 2c^2 + c - 3.$$

For $c \geq 2$, the right-hand side is at most $-(2c-3)(c-1) < 0$, using $n \geq 2c$. For $c = 1$ it forces $ab = 0$, giving a reflection or its negative. For $c = 0$ it gives $(a-2)(b-2) = 1$, whose only solution with $n \geq 3$ is $(a, b) = (3, 3)$ in B_6. That class has size $\binom{6}{3}/2 = 10$ in $W/Z(W)$, whereas the long-reflection class has size 30. An automorphism cannot exchange them. For a short reflection the value is $n^2 - 2n + 2$, and the equation is

$$ab = n(1-2c) + 2c^2 + c - 1.$$

Its right-hand side is negative for $c \geq 1$, $n \geq 3$, $n \geq 2c$ (for $c = 1$ it is $2 - n$). For $c = 0$ it reduces to $(a-1)(b-1) = 0$, giving the short reflections and their negatives. The two reflection character values differ when $n \geq 3$.

For D_n the reflection value is $(n-2)(n-3) + 2$, and the equation becomes

$$ab = 2n(1-c) + 2c^2 + 2c - 4.$$

For $c = 2$ its right-hand side is $8 - 2n < 0$; for $c \geq 3$ it is at most $-2(c-1)(c-2) < 0$. For $c = 1$ we again get $ab = 0$. For $c = 0$ we get $(a-2)(b-2) = 0$. The last possibility is represented, up to the central sign when available, by negating two coordinates. Its class has size $\binom{n}{2}$, whereas the reflection class has size $n(n-1)$. For $n \geq 5$ passing to $W/Z(W)$ does not halve either size: a lift and its negative have different signed cycle types whenever both belong to W. This excludes the extra class and proves the lemma. □

Proposition 6.4. *The quotients Γ_{A_n} $(n \geq 2)$, Γ_{B_n} $(n \geq 3)$, and Γ_{D_n} $(n \geq 4)$ have property R_∞.*

Proof. Let D be the linear part of a quotient automorphism and θ its holonomy automorphism. It preserves $\chi_{\mathcal{T}}$, and every permutation of $\mathcal{T}$ preserves the translation lattice described above.

Type A_n. The proof of Proposition 5.1 applies directly: the exceptional outer automorphism of S_6 does not preserve $\chi_{\mathcal{T}}$, so θ is inner, and the standard constituent $S^{(n,1)}$ supplies the two signs.

Types B_n, D_n, $n \geq 5$ in type D. By Lemma 6.3, θ preserves reflections. For D_n, Lemma B.1 makes it inner up to a diagram automorphism. The same root argument works for B_n, as follows. Choose a root β_i for each image of a simple reflection, with the same length as the corresponding simple root α_i; length is preserved by the lemma. Orders of products of two distinct reflections agree in W and $W/Z(W)$, because their plane action fixes a nonzero complement and hence no nontrivial power can equal $-I$. The Coxeter product orders therefore determine $|(\beta_i, \beta_j)|$. Choose signs along the diagram tree to match all simple-root inner products. The isometry $\alpha_i \mapsto \beta_i$ maps the root system into itself, hence onto itself, since the images of simple reflections generate a subgroup of W. In type B_n every such isometry is a signed permutation (it permutes the short roots), so it belongs to W. Thus θ is inner in type B_n. The U^- witnesses of Proposition 5.2, including its diagram-involution case for D_n, now apply through Proposition 6.1. This includes B_3 and B_4.

Type D_4. The frame constituent U of Proposition 5.4 is the unique rational irreducible constituent of dimension two. Indeed, the sign-fixed part is the S_4-module on six pairs, with dimensions $1, 3, 2$. On U^-, the three distinct nontrivial sign characters are permuted transitively, so every irreducible constituent has dimension divisible by three. Hence $DU = U$. The holonomy image on U is S_3. Every automorphism of S_3 is inner, so $D|_U = c\rho_U(g)$ for some g and $c \in \mathbb{Q}^\times$. Both operators

preserve $U \cap L$, forcing $c = \pm 1$. A reflection of this S_3 has eigenvalues $1, -1$, giving the required fixed vector after a holonomy twist. □

6.4. **The odd-dimensional alternative.** The multiplicity-one argument of Dekimpe–De Rock–Penninckx [8, Thm. 4.7], combined with Hiss–Lutowski [25, 26], provides an alternative for five of the seven types. We give the reduction explicitly, since the flat-manifold statement in [8] assumes torsion-freeness, whereas our quotients may have torsion.

Suppose V has a nontrivial good constituent U of odd dimension d, and no other irreducible complex constituent has dimension d. Every $D \in N_{GL(L)}(G)$ preserves U: it permutes isotypic components and preserves their irreducible dimensions. Moreover $N_{GL(U\cap L)}(G|_U)$ is finite. Indeed, conjugation maps this normaliser to the finite group $\mathrm{Aut}(G|_U)$, with kernel $\{\pm I\}$ by absolute irreducibility and integrality. Hence $D|_U$ has finite order. The Hiss–Lutowski theorem supplies $g \in G$ with $gD|_U$ having eigenvalue one. The determinant criterion [11, Thm. 3.3], valid with torsion, proves R_∞. The uniqueness of the constituent also verifies the non-conjugacy hypothesis in [8, Thm. 4.7].

type	odd dimension d	explicit witnesses	odd-dimensional route
D_5	5	applies	applies
E_6	15	applies	applies
E_7	27	applies	applies
E_8	35	applies	applies
H_4	25	applies	applies
F_4, H_3	—	applies	not supplied by this criterion

Here are the constituent checks. For E_n, Proposition 4.2 gives the complete decomposition. For D_5, the sums $e_{[e_i+e_j]} + e_{[e_i-e_j]}$ form the permutation module on the 2-subsets of five letters, with irreducible dimensions $1, 4, 5$ by Young's rule; its complement is the good 10-dimensional U^- of Proposition 5.2. For H_4, the 5-dimensional irreducible representation of A_5 is defined over $\mathbb{Q}$: it is the augmentation module of the action on the six Sylow 5-subgroups. This action is doubly transitive, so the augmentation module is absolutely irreducible. Its matrix coefficients give a rational good constituent of dimension 25, by the argument of Proposition 4.6(a); the remaining complex dimensions are $1, 9, 9, 16$. Thus the required odd constituent is unique in each case. For H_3 the odd nontrivial constituent has multiplicity two, and for F_4 the explicit proof uses the exchanged 2-dimensional constituents. The table concerns this particular odd-dimensional route, not every possible application of [25, 26].

6.5. **Compatibility with the extension.** An infinite integral normaliser alone does not decide R_∞. The extension determines which of its elements actually occur as linear parts. We record the standard compatibility test (cf. [11, Algorithm 1]) in notation useful for the counterexample.

Proposition 6.5. *Write* $\Gamma = \bigcup_{g\in G}(a_g + L, g) \le V_{\mathbb{R}} \rtimes G$, *with* $a_1 = 0$, *and let* $D \in N_{GL(L)}(G)$ *and* $\theta(g) = DgD^{-1}$. *Conjugation by* (d, D) *is an automorphism of* Γ *if and only if*

$$Da_g + (I - \theta(g))d - a_{\theta(g)} \in L \qquad (g \in G). \tag{1}$$

For such an automorphism, R *is finite if and only if* $\det(I - gD) \neq 0$ *for every* $g \in G$.

Proof. Affine conjugation sends (a_g, g) to $(Da_g + (I - \theta(g))d, \theta(g))$. Condition (1) is exactly membership in the required coset. Since $DL = L$ and θ is onto, these cosets map onto all of Γ. The last assertion is [11, Thm. 3.3]. It is enough to check membership on a set of generators of Γ, together with $DL = L$ and surjectivity on G. □

Equivalently, D must preserve the extension class with the indicated change of holonomy action. This condition is separate from avoiding eigenvalue one in all the matrices gD.

6.6. **The dihedral classification.**

Theorem 6.6. *For $m \geq 3$, the quotient $\Gamma_{I_2(m)}$ has R_∞ if and only if $4 \mid m$, or m is odd and $3 \mid m$.*

Proof. On the basis e_j, $j \in \mathbb{Z}/m\mathbb{Z}$, put $Ce_j = e_{j+1}$ and $Se_j = e_{-j}$. The actual quotient is generated affinely by (e_0, S) and (e_1, C^2S), with holonomy $G = \langle C^2, S\rangle$. Its translation lattice is $2\mathbb{Z}^m$ for odd m and $2\mathbb{Z}^m + \mathbb{Z}\mathbf{1}$ for even m.

Odd m divisible by three. The rational decomposition in Remark 5.5 has a unique 2-dimensional rational irreducible constituent U_3, and its holonomy image is S_3. Every rational normaliser preserves this constituent. The S_3 argument in the D_4 case of Proposition 6.4 gives a fixed vector after a holonomy twist, so R_∞ holds.

m divisible by four. Write $m = 2q$, with q even. The only nontrivial rational 1-dimensional constituents are the multiplicity-one lines spanned by

$$u_0 = \sum_{j=0}^{q-1} (-1)^j e_{2j}, \qquad u_1 = \sum_{j=0}^{q-1} (-1)^j e_{2j+1}.$$

Indeed, on each parity orbit C^2 is a q-cycle, so its 1- and (-1)-eigenspaces are lines; the 1-eigenspaces are trivial G-modules. Both displayed lines have character -1 on C^2, and their characters on S are 1 and -1, respectively. Any $D \in N_{GL(L)}(G)$ preserves or swaps these lines. If it preserves them, it acts on each by ± 1, and a holonomy twist gives eigenvalue one. If it swaps them, D^2 acts by a common scalar $c = \pm 1$, by preservation of the one-dimensional lattices. On their sum, $(gD)^2$ acts by $c\chi_0(g)\chi_1(g)$. Choose $g = 1$ or S to make this scalar 1. Then $v + gDv$ is a nonzero fixed vector for any nonzero v on the first line.

Odd m not divisible by three. Put $D = I - C - C^{-1}$ and $d = e_0 + e_1$. The identity $(I + C)(I - C + C^2) = I + C^3$ shows $\det(I - C + C^2) = 1$: both C and C^3 are m-cycles and $\det(I + C) = \det(I + C^3) = 2$ for odd m. Since $D = -C^{-1}(I - C + C^2)$, it follows that $\det D = -1$. Thus D preserves $L = 2\mathbb{Z}^m$ and commutes with G. Moreover,

$$(D - I)e_0 + (I - S)d = -2e_{m-1}, \qquad (D - I)e_1 + (I - C^2S)d = -2e_2.$$

Proposition 6.5 therefore gives an automorphism of the actual quotient. The eigenvalues of D are $1 - 2\cos(2\pi j/m)$. The only one of absolute value one is -1 on the constant line: an eigenvalue 1 would require $4 \mid m$. As D commutes with the finite-order operators $g \in G$, their restrictions to its eigenspaces have roots of unity as eigenvalues. Hence gD cannot have eigenvalue one off the constant line, and on that line it acts by -1. All the determinants $\det(I - gD)$ are nonzero, proving finite R.

$m \equiv 2 \pmod 4$. Write $m = 2q$ with q odd. The two parity orbits of reflections are G-equivariantly identified by $j \mapsto j + q$. In the ordered pairs (e_j, e_{j+q}), j even, let

$$D = I_q \otimes \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}, \qquad d = 0.$$

This operator commutes with G, is integral unimodular, and is congruent to I modulo two. The same is true of D^{-1}, so both preserve $L = 2\mathbb{Z}^m + \mathbb{Z}\mathbf{1}$. The generator corrections $(D-I)e_0, (D-I)e_1$ lie in $2\mathbb{Z}^m$, proving extension compatibility. The eigenvalues of D are $3 \pm 2\sqrt{2}$, neither of modulus one. As in the preceding case, no gD has eigenvalue one. This supplies an automorphism with finite R and completes the classification. □

Proof of Theorem B. Theorem 6.2 and Proposition 6.4 cover every irreducible spherical type of rank at least three. Theorem 6.6 covers rank two, including $A_2 = I_2(3)$ and $B_2 = I_2(4)$. □

6.7. **A dihedral counterexample.**

Proposition 6.7 (Proposition C)**.** *The group* $\Gamma = A(I_2(5))/[P,P]$ *has an automorphism* Φ *with* $R(\Phi) = 18$.

Proof. Index the reflections by $j \in \mathbb{Z}/5\mathbb{Z}$, with the two simple reflections indexed by $0, 1$. On the reflection lattice set

$$Ce_j = e_{j+1}, \qquad Se_j = e_{-j}, \qquad G = \langle C, S\rangle.$$

The two simple reflections act as S and C^2S. By Proposition 2.4, the actual quotient is

$$\Gamma = \langle (e_0, S), (e_1, C^2S)\rangle \leq \mathbb{Z}^5 \rtimes G, \qquad L = 2\mathbb{Z}^5.$$

Here $Z(W) = 1$, so L is the full translation lattice. Put

$$D = I - C - C^{-1}, \qquad d = e_0 + e_1, \qquad \Phi(v, g) = (Dv + (I - g)d, g).$$

We have $D^{-1} = I - C^2 - C^{-2}$ and D commutes with G, so $DL = L$. For the two generators,

$$(D - I)e_0 + (I - S)d = -2e_4, \qquad (D - I)e_1 + (I - C^2S)d = -2e_2.$$

These corrections lie in L. Thus conjugation by (d, D) preserves Γ and is onto, by Proposition 6.5. This checks the extension as well as the integral lattice.

The characteristic polynomial of D is $(x+1)(x^2-3x+1)^2$, and direct integer determinants give

g	1	C^k $(1 \leq k \leq 4)$	C^kS $(0 \leq k \leq 4)$
$\det(I - gD)$	2	22	-10.

In particular all determinants are nonzero, which already proves $R(\Phi) < \infty$.

For completeness we count the classes without a torsion-free assumption. Since Φ induces the identity on G, twisted classes project to ordinary conjugacy classes of G, represented by $1, C, C^2, S$. Choose lifts $(a_g, g) \in \Gamma$. Twisted conjugation by translations identifies the fibre above g with

$$Q_g = \mathbb{Z}^5/(I - gD)\mathbb{Z}^5, \qquad x \longleftrightarrow (a_g + 2x, g).$$

The remaining identifications in this fibre are precisely the action of the centraliser $C_G(g)$. For a lift (b_h, h) with $hg = gh$, that action is

$$x \longmapsto hx + \tfrac{1}{2}\big(b_h + ha_g - gDb_h - g(I - h)d - a_g\big) \pmod{(I - gD)\mathbb{Z}^5}. \tag{2}$$

The following table gives lifts, cyclic coordinates k on Q_g, and the induced actions of centraliser generators. All vectors are ordered from 0 to 4.

g	a_g	Q_g	centraliser action	orbits
1	$(0,0,0,0,0)$	$\mathbb{Z}/2$	$C : k \mapsto k$, $S : k \mapsto k+1$	1
C	$(1,0,1,1,1)$	$\mathbb{Z}/22$	$C : k \mapsto 5k+6$	6
C^2	$(0,1,1,0,0)$	$\mathbb{Z}/22$	$C : k \mapsto 3k+4$	6
S	$(1,0,0,0,0)$	$\mathbb{Z}/10$	$S : k \mapsto 1-k$	5

Here $a_{C^2} = e_1 + e_2$ comes from the product of the second generator and the first, and a_C is the parity representative of its cube. The cyclic coordinates are $k = \ell_g x$ modulo the stated order, with

$$\ell_1 = (1,1,1,1,1), \quad \ell_C = (1,5,3,15,9), \quad \ell_{C^2} = (1,3,9,5,15), \quad \ell_S = (5,1,7,3,9).$$

Each functional is onto and annihilates $(I - gD)\mathbb{Z}^5$ modulo $|\det(I - gD)|$, so it identifies Q_g with the indicated cyclic group. Substitution in (2) gives the table. Each order-five action on $\mathbb{Z}/22$ has two fixed points and four orbits of length five; the involution $k \mapsto 1-k$ on $\mathbb{Z}/10$ has no fixed points. Consequently

$$R(\Phi) = 1 + 6 + 6 + 5 = 18.$$

□

Remark 6.8 (Torsion and the determinant average)**.** The element $((-1,0,-1,1,1),C)$ belongs to Γ and has order five: its translation vector differs from a_C by an element of $2\mathbb{Z}^5$, and its five translates sum to zero. Thus the Bieberbach averaging formula [11, Thm. 3.4] cannot be invoked here. In fact its determinant average is $(2+4\cdot 22+5\cdot 10)/10 = 14$, whereas $R(\Phi) = 18$. The exact checker `code/indep/dihedral_quotient.py` verifies the preceding calculation and independently enumerates all 140 translation-twisted classes: there are 13 holonomy orbits of size 10 and 5 of size 2.

Remark 6.9 (The representation-theoretic distinction)**.** For $I_2(5)$, $\mathbb{Q}^5 = \mathbb{Q}\mathbf{1} \oplus U$, where U is rationally irreducible but splits into two conjugate 2-dimensional constituents over $\mathbb{Q}(\sqrt{5})$. Its commuting algebra is $\mathbb{Q}(\sqrt{5})$. The operator D acts by -1 on the trivial line and by the conjugate units $(3\pm\sqrt{5})/2$ on those two constituents. These units have infinite order. The displayed generator corrections are what make this particular unit compatible with the extension; the existence of units alone would not establish Proposition C.

Further question. For the negative dihedral cases in Theorem B, determine the Reidemeister spectrum $\{R(\Phi) : \Phi \in \mathrm{Aut}(\Gamma_{I_2(m)})\}$. Proposition C shows that 18 belongs to this spectrum for $m = 5$. The extension-compatible integral normaliser and the torsion-sensitive class count both matter for this more detailed question.

Appendix A. Computational certificates and cross-checks

Programs, commands and expected outputs are listed in `code/README.md` of the accompanying repository. We distinguish *proof certificates* from *corroboration*.

- Proof certificates are finite computations that the written proofs rely on. They use exact arithmetic over $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{5})$, including exact rational row reduction for ranks, and the programs assert the expected values (a nonzero exit status on any discrepancy).
- Corroboration consists of independent recomputations, some of them numerical.

Coordinates. For E_8 we use Bourbaki's simple roots in $\mathbb{R}^8$ with basis $e_1,\dots,e_8$:

$$\alpha_1 = \tfrac{1}{2}(e_1+e_8) - \tfrac{1}{2}(e_2+\cdots+e_7), \qquad \alpha_2 = e_1+e_2, \qquad \alpha_k = e_{k-1}-e_{k-2} \quad (3\le k\le 8).$$

E_7 and E_6 use $\alpha_1,\dots,\alpha_7$, resp. $\alpha_1,\dots,\alpha_6$, in $\mathbb{R}^8$. The other types:

- F_4: $e_2-e_3,\ e_3-e_4,\ e_4,\ \frac{1}{2}(e_1-e_2-e_3-e_4)$;
- A_n, B_n, D_n: standard simple roots;
- H_3: roots $\pm 2e_i$ and the even permutations of $(\pm\tau, \pm 1, \pm\tau^{-1})$, with $\tau = \frac{1+\sqrt{5}}{2}$;
- H_4: roots $\pm 2e_i$, $(\pm 1,\pm 1,\pm 1,\pm 1)$ and the even permutations of $(\pm\tau,\pm 1,\pm\tau^{-1},0)$.

Proof certificates.

(1) Proposition 4.2: perpendicular root counts and their exact ranks (used to identify the subsystems), together with the SRG parameters, the three orbitals and the eigenspace dimensions.

(2) Propositions 4.3–4.6 and Lemma 4.5:
   - frames, and character norms $\langle\chi,\chi\rangle = \langle\chi,\chi\tau\rangle = 1$;
   - the images of W on frames (A_5, S_3);
   - the action of π_Σ, π_L, π_Ψ, with eigenspace dimensions $(3,1)$, $(1,1)$, $(10,6)$;
   - the closure of I, the reflection formula, and $\rho(W) = \{x \mapsto ax^{\pm 1}b\}$;
   - $K \cong A_5$: $|K| = 60$, 15 involutions, 5 Klein four-subgroups, and conjugation is an injective homomorphism into A_5; also $Z(K) = 1$;
   - $\Psi(I) = I$, Ψ multiplicative, and the real-part computation in Proposition 4.6(b).

(3) Propositions 5.2 and 5.4:
   - the weight argument for B_3–B_6 and D_4–D_7, including its failure for D_4;

- the operator J and the frame module of D_4;
- witnesses for all six graph automorphisms of D_4.

(4) Theorem B uses the full holonomy-automorphism certificates in Appendix B. Proposition C has the direct calculation of Section 6.7; its independent exact checker verifies the extension, all determinants, the cyclic quotient table, and the full twisted-class count.

Corroboration.

(5) Earlier, independent certificates for the six exceptional types:
- commutant certificates;
- character certificates written as fresh code;
- witnesses for abstractly computed outer automorphisms.

They give the same dimensions $15, 20$; $27, 35$; $35, 84$; 4; $2+2$; $16, 25$.

(6) The criterion (for automorphisms preserving $\chi_{\mathcal{T}}$) checked uniformly for A_2–A_7, B_3–B_7, D_4–D_7, odd $I_2(m)$ with $m \leq 21$, and the exceptional types other than E_8.

(7) Independent checkers written by the internal AI referees. The one for D_n, F_4, H_3 and parts of the one for H_4 use numerical linear algebra, so they are numerical corroboration only; items (1)–(3) above are their exact counterparts.

(8) The uniform arguments in Lemma 6.3 and Theorem 6.6 have an independent finite cross-check in `code/indep/quotient_families.py`. It counts fixed root lines directly, checks the exceptional character coincidences, and verifies the positive constituents or the affine maps and all adjusted determinants in the dihedral cases. The unbounded statements are proved in the text, not inferred from these finite runs.

Appendix B. Automorphisms of the holonomy groups

The Artin-group argument uses automorphisms of $\rho(W)$ induced from $\mathrm{Aut}(W)$. For the seven cases of Theorem 6.2, we control all holonomy automorphisms, a sufficient strengthening of the character-preserving restriction. The following computations are proof certificates for these cases of Theorem B, as well as corroboration of [22, Thm. 31]:

- $\mathrm{Out}(\rho(W)) = 1$ for E_6, E_7, E_8;
- $|\mathrm{Out}(\rho(W))| = 2$ for H_3, H_4, F_4.

We also record $\mathrm{Out}(\rho(W(D_5))) = 1$. These statements were obtained as follows.

- *Involution classes and Coxeter tuples*, by exact enumeration, for $H_3, H_4, F_4, D_5, E_6, E_7$. No involution class other than the reflection classes has a reflection-class size, so every automorphism preserves $\rho(\mathcal{T})$; then the Inn-orbits of generating Coxeter tuples of reflections are counted.
- *Brute force over generator images* (all automorphisms, without assuming that reflections are preserved) for E_6, H_3, H_4, F_4, D_5, and independently by the internal AI referee for H_3, F_4. This program was *not* run for E_7.
- For E_7 and E_8, the reflection-preserving stage was re-verified with Schreier–Sims stabiliser chains, and it also follows from Lemma B.1 below. For E_7 the reflection-preservation stage therefore rests on the single exact enumeration above.
- For E_8, Proposition B.2 below gives a proof by hand.

Lemma B.1. *Let Φ be an irreducible simply-laced root system of rank ≥ 3 (so its diagram is a tree). Then every $\theta \in \mathrm{Aut}(\rho(W))$ with $\theta\rho(\mathcal{T}) = \rho(\mathcal{T})$ is inner up to a graph automorphism.*

Proof. Write $\theta\rho(s_i) = \rho(s_{\beta_i})$; this uses that ρ is injective on $\mathcal{T}$. Orders of products agree in W and in $\rho(W)$. Indeed, for roots $\beta \neq \pm\gamma$ the element $s_\beta s_\gamma$ is a plane rotation fixing a complement of dimension ≥ 1, so no nontrivial power of it lies in $Z(W) \subseteq \{\pm 1\}$; in particular $s_i s_j \neq -1$. Hence $(\beta_i, \beta_j) = 0$ or ± 1 according to the diagram. Choose the signs along the tree. Then $g\colon \alpha_i \mapsto \beta_i$ is an isometry, and $g\Phi = (gWg^{-1})\{\beta_i\} \subseteq \Phi$. So $g\{\alpha_i\}$ is a simple system. As W is simply transitive on

simple systems, $g = w\gamma$ with $w \in W$ and γ an isometry permuting $\{\alpha_i\}$, i.e. $g \in W \rtimes \mathrm{Gr} = \mathrm{Aut}(\Phi)$ [37]. Compare [22, Thm. 18, Cor. 19]. □

Proposition B.2. *Every automorphism of $W(E_8)/\{\pm 1\}$ is inner.*

Proof. $G^+ = W^+/\{\pm 1\}$ is the commutator subgroup, of index 2. An involution outside G^+ lifts to w with $w^2 = 1$ and $\det w = -1$; if $w^2 = -1$ then $\det w = 1$. By [6, Lemma 5] (provable via Steinberg's theorem), w is a product of $k = \dim\ker(w+1)$ reflections in mutually orthogonal roots, and k is odd. Replacing w by $-w$, we may take $k \in \{1, 3\}$. For $k = 3$, the (-1)-eigenspace of w is spanned by the three roots, and contains only the 6 roots $\pm\beta_i$: a root $\sum c_i\beta_i$ has $2c_i = (\cdot, \beta_i) \in \{0, \pm 1, \pm 2\}$ and $\sum c_i^2 = 1$. So w determines the triple, and $-w$ (with $k = 5$) is not of this form.

The products for $k = 3$ form one class of size $240 \cdot 126 \cdot 60/48 = 37800$, by transitivity along $E_8 \supset E_7 \supset D_6$. So the reflections form the unique class of size 120 outside G^+. Hence every automorphism preserves $\rho(\mathcal{T})$, and Lemma B.1 applies. □

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